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volesti: A C++ library for sampling and volume computation on convex bodies

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Summary

Sampling from (constrained) high-dimensional distributions and volume approximation of convex bodies are fundamental operations that appear in optimization, finance, engineering, artificial intelligence, and machine learning. We present `volesti`, a C++ library that delivers efficient implementations of state-of-the-art, mainly randomized, algorithms to sample from general logconcave distributions. Based on these routines, we can estimate the volume of convex bodies in high dimensions, round them, and compute multidimensional integrals over them. The backbone of our library consists of Monte Carlo algorithms, which are randomized algorithms, the output of which can be incorrect with (usually very small) error probability; thus, we also provide several high-dimensional statistical tests to certify and verify the output.

The focus of `volesti`' is scalability in high dimensions, that, depending on the problem at hand, could range from hundreds to thousands of dimensions. Another novelty is the ability to handle a variety of different inputs for the constrained support of the various distributions. `volesti` supports three different types of polyhedra (Ziegler, 1995), spectrahedra (Ramana & Goldman, 1999) and general non-linear convex objects.

`volesti` relies on `Eigen` library (Guennebaud et al., 2010) for linear algebra but also supports `MKL` optimizations (*Intel Math Kernel Library (Intel MKL)*, 2024). There are R (Chalkis & Fisikopoulos, 2021) and Python (Chalkis, Fisikopoulos, Tsigaridas, et al., 2023) interfaces available.

Statement of need

High-dimensional sampling from multivariate distributions with Markov Chain Monte Carlo (MCMC) algorithms is a fundamental problem with many applications in science and engineering (Genz & Bretz, 2009; Iyengar, 1988; Schellenberger & Palsson, 2009; Somerville, 1998). In particular, multivariate integration over a convex set as well as the volume approximation of convex sets have garnered significant attention from theorists and engineers over the last decades. Nevertheless, these problems are computationally hard for general dimensions (Dyer & Frieze, 1988). MCMC algorithms made remarkable progress and their use allowed us to efficiently tackle the problems of sampling and volume estimation of convex bodies in theory, by the introduction of (rigorous) theoretical guarantees (Chen et al., 2018; Lee & Vempala, 2018; Mangoubi & Vishnoi, 2019). Unfortunately, these theoretical guarantees of the MCMC algorithms do not extend in a straightforward manner to efficient implementations able to attack problems coming from real-life computations. Therefore, we witnessed the birth of efficient in practice MCMC algorithm that relax the theoretical guarantees and employ new algorithmic and statistical techniques to be amenable to efficient implementations. Remarkably,

these algorithms, and the corresponding implementations, also meet the requirements for high accuracy results (Chalkis, Emiris, et al., 2023; Cousins & Vempala, 2016; Emiris & Fisikopoulos, 2014; Kook et al., 2022); however several existing published methods are available as part of propertiary packages (MATLAB) (Cousins & Vempala, 2016; Kook et al., 2022).

Our open-source package – volesti – offers all of the aforementioned functionality, together with the support of sampling from general log-concave densities Chalkis, Fisikopoulos, Papachristou, et al. (2023), and uniform sampling from spectrahedra Chalkis et al. (2022).

Our implementation supports: 1. support various sampling techniques based on geometric walks, roughly speaking these are a continuous version of MCMC algorithms, such as Billard walk, Hamiltonian walk and others, 2. give the user the ability to sample from various distributions, like uniform, exponential, Gaussian, and general log-concave densities, 3. allows to consider the distributions constrained in various convex domains, such as hypercubes, zonotopes, general polytopes (defined either as a set of linear inequalities or as a convex hull of a pointset), spectrahedra (feasible sets of semidefinite programs), and, 4. can perform volume computations, integration, and solve problems from real life applications in very high dimensions.

Impact

`volesti` has been used extensively in various research and engineering projects coauthored by the authors of this paper. In particular, for the problem of sampling the flux space of metabolic networks we were able to sample from the most complicated human metabolic network accessible today, Recon3D (Chalkis et al., 2021), used to model financial crises (Calès et al., 2023), to detect low volatility anomalies in stock markets (Bachelard et al., 2023), to introduce randomized control in asset pricing and portfolio performance evaluation (Bachelard et al., 2024)), but also to sample from (and compute the volume of) spectrahedra (Chalkis et al., 2022), the feasible regions of semidefinite programs.

Even more, `volesti` has been used by other research teams in conducting research in electric power systems (Venzke et al., 2021), for problems in probabilistic inference (Spallitta et al., 2024), to perform resource analysis on programs (Pham, 2024); but also for more theoretical and mathematical challenges, like the computation of topological invariants (Maria & Rouillé, 2021), and persistent homology (Vejdemo-Johansson & Mukherjee, 2022).

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